Exercise Class 2 — Production (MWG Ch. 5)

Advanced Microeconomics 1 · Monday 19 October, 16:00–18:00

Format (120 min). 60 min: assigned students present Problems 1–3, 5 and 6 at the board (assignments posted Thursday 15 October). 30 min: instructor works the hard item (Problem 4). 30 min: timed prelim-style question (Problem 7), closed book, ungraded, self-marked against the sketch afterwards.

Also today: the two-page *course synopsis* is distributed — the map of the course plus the ten derivations you should be able to produce cold. Bring it to EC3.

Part I — Problems

Problem 1 Production set structure

(a) Show: if Y is convex and $0 \in Y$, then Y exhibits nonincreasing returns to scale.

(b) Show: if Y is additive and exhibits nonincreasing returns to scale, then Y is a convex cone. (Hint: build rational scalings from replication and scaling-down; use closedness for the irrational ones.)

(c) Interpret (b) economically: what does free entry/replication do to decreasing returns at the industry level, and what does that imply for the shape of long-run industry supply?

Problem 2 The Cobb–Douglas firm, end to end

Let $f(z_1, z_2) = z_1^{\alpha} z_2^{\beta}$ with $\alpha, \beta > 0$, $\alpha + \beta < 1$; input prices $w \gg 0$, output price p.

(a) Derive the conditional factor demands $z(w, q)$ and cost function $c(w, q)$.

(b) Verify Shephard's lemma and the concavity of c in w (check the Hessian is NSD, or argue via the envelope-of-lines).

(c) Derive supply $q(p, w)$ from $p = \partial c/\partial q$, then the unconditional factor demands and $\pi(p, w)$.

(d) Verify Hotelling's lemma in both arguments: $\partial \pi/\partial p = q$ and $\partial \pi/\partial w_\ell = -z_\ell$.

(e) What happens to each object as $\alpha + \beta \to 1$? Reconcile with the CRS pathology of the PMP.

Problem 3 The law of supply, and why no compensation

(a) Prove: for any p, p' and profit-maximizing $y \in y(p)$, $y' \in y(p')$, $(p - p') \cdot (y - y') \geq 0$.

(b) State the consumer-side analogue (compensated law of demand) and identify precisely which feature of the consumer's problem makes compensation necessary there and unnecessary here.

(c) Deduce from (a): a competitive firm's output never falls when its output price rises, and its demand for an input never rises when that input's price rises. No differentiability, no convexity.

Problem 4 Markups from the cost side — and their identification

A firm produces $Q = F(V, K)$, chooses the flexible input V (price P_V) each period taking P_V as given, and may have arbitrary market power in its output market (price P).

(a) From cost minimization, derive $\mu \equiv P/MC = \theta_V \cdot \dfrac{PQ}{P_V V}$ where $\theta_V = \dfrac{\partial F}{\partial V}\dfrac{V}{Q}$.

(b) Suppose demand is CES with elasticity $\varepsilon > 1$ and the firm sets price optimally: show $\mu = \varepsilon/(\varepsilon - 1)$ and that the revenue share of V is $s_V = \theta_V/\mu$.

(c) An econometrician observes *revenue* $R = PQ$, not quantity. Show that the elasticity of revenue with respect to V is $\theta_V^R = \theta_V\,(1 - 1/\varepsilon) = \theta_V/\mu$.

(d) Conclude: $\hat{\mu} = \theta_V^R/s_V = 1$ identically, regardless of the true markup. State in one sentence what data or assumption breaks this impossibility (Bond–Hashemi–Kaplan–Zoch 2021).

Problem 5 Gorman form: the price of the representative consumer

Consumers $i = 1, \ldots, I$ have indirect utilities of the Gorman form

$$v_i(p, w_i) = a_i(p) + b(p)\,w_i, \qquad b(p) > 0 \text{ common to all } i.$$

(a) Apply Roy's identity to derive $x_i(p, w_i)$ and show individual demand is affine in own wealth with a wealth-slope $-\nabla_p b(p)/b(p)$ that is *identical across consumers.*

(b) Show aggregate demand $\sum_i x_i(p, w_i)$ depends on $(w_1, \ldots, w_I)$ only through $\sum_i w_i$.

(c) Exhibit a representative consumer: an indirect utility $V(p, W)$ with $W = \sum_i w_i$ whose Roy demand equals the aggregate.

(d) Name two familiar preference classes in Gorman form (identical homothetic; quasilinear with common numeraire), and explain in two sentences why the failure of Gorman form in the data is the reason heterogeneous-agent macroeconomics exists.

Problem 6 Efficiency and supporting prices

(a) Prove MWG 5.F.1: if $y \in y(p)$ for some $p \gg 0$, then y is efficient.

(b) Give an example (a picture suffices) of an efficient production plan in a *convex* Y that is profit-maximizing only for a price vector with a *zero* component; explain why $p \gg 0$ cannot be guaranteed in Prop. 5.F.2.

(c) Give a nonconvex Y (picture) with an efficient point that is profit-maximizing for *no* price vector. Which step of the separating-hyperplane proof dies?

(d) In two sentences: how do Hsieh–Klenow (2009) convert the *failure* of common supporting prices into a measurable object, and what is that object?

Problem 7 Timed prelim question (30 minutes, closed book)

A price-taking firm in input markets has cost function

$$c(w_1, w_2, q) \;=\; 2\,q\,\sqrt{w_1 w_2}.$$

(a) Verify that c has the properties a cost function must have: homogeneous of degree one, nondecreasing, and concave in w; nondecreasing in q.

(b) Recover the conditional factor demands $z(w, q)$ by Shephard's lemma, and identify the underlying technology.

(c) The firm sells its output at price p, taking p as given. Derive the supply correspondence and the profit function. Comment on the returns to scale.

(d) Now suppose the firm has market power and you observe: expenditure on inputs equal to 60, revenue equal to 90, and you are told the technology is as above (so the output elasticity of the variable input bundle is 1). Compute the markup μ. State one assumption this calculation leans on that you cannot verify from these data.